MTH107

Timed Online Assignment – January Semester 2022

Calculus I

Tuesday, 17 May 2022 **1:00 pm – 3:00 pm**

Time allowed: 2 hours

INSTRUCTIONS TO STUDENTS:

1. This Timed Online Assignment (TOA) contains **FIVE (5)** questions and comprises **FIVE (5)** pages (including cover page).

2. You must answer **ALL** questions.

3. If you have any queries about a question, or believe there is an error in the question, briefly explain your understanding and assumptions about that question before attempting it.

4. You MUST **submit your answers via Canvas (similar to TMA submission) at the end time of this TOA** (as stated on this cover page). The 15 minutes grace period as shown on Canvas is strictly meant for technical issues encountered during submission. Thereafter, you will not be able to submit your answers and you will be considered as having withdrawn from the course. **No appeal will be allowed.**

5. You are to include the following particulars in your submission: TOA Course Code, your Full Name, and SUSS PI Number. Name your submission file as TOACourseCode_FullName_StudentPI. Use underscore and NOT Space. *Example: TOAXYZ_RaphaelLee_T1923161 (omit D/O, S/O).*

6. Your submission should consist of only one file and **must not exceed 500MB** in size. The file must be a PDF or Microsoft Word file saved in .doc or .docx format. All answers are to be typed. Flowcharts and graphs may be scanned or photographed and embedded in the PDF or Word file provided it does not exceed the file size limit of 500MB.

7. For answers which cannot be typed and require to be hand-written, ensure that question number is clearly stated on each page. All uploaded hand-written answers must be clear, readable and complete. Marks will not be awarded for un-readable or incomplete images.

8. To prevent plagiarism and collusion, your submission will be reviewed by Turnitin. The Turnitin report will only be made available to the marker and you will not be able to view it.

9. The University takes plagiarism and collusion seriously, and your Turnitin report will be examined thoroughly as part of the marking process.

10. Marks will only be awarded if FULL working is shown.

Answer all questions. (Total 100 marks)

Question 1

(a) Determine the following limit. Show your steps clearly.

$$\lim_{x \to 2} \frac{x^5 - 32}{x - 2}.$$

(6 marks)

(b) Determine the following limit. Show your steps clearly.

$$\lim_{x \to 3} \frac{x^2 - x - 6}{x - 3}.$$

(7 marks)

(c) Show that there exists a positive real root for the equation $\cos x = x^2 - 1$. Justify your answer.

(7 marks)

Question 2

(a) Let f be the function defined on $\mathbb{R}$ by

$$f(x) = \begin{cases} x^4 e^{\sin\left(\frac{1}{x}\right)}, & if\ x \neq 0 \\ 0, & if\ x = 0 \end{cases}.$$

Determine whether f is continuous at 0.

(7 marks)

(b) Let f be the function defined by

$$f(x) = \begin{cases} \dfrac{ax}{|x|}, & if\ x \neq 0 \\ b, & if\ x = 0 \end{cases}.$$

Find all possible values of $a, b \in \mathbb{R}$ such that f is continuous at 0. Justify your answer.

(7 marks)

(c) Let $f(x) = |x \cos x|$. Is f differentiable at 0? If yes, determine the value of $f'(0)$. Justify your answer.

(6 marks)

Question 3

(a) Determine the limit

$$\lim_{x \to 0} \frac{\cos x - 1}{x^2}.$$

Show your steps clearly.

(10 marks)

(b) Show that

$$|\ln x - \ln y| \leq 4|x - y|\,,$$

for all $\frac{1}{4} \leq x < y \leq 4$.

(10 marks)

Question 4

(a) **<u>Without using L'Hopital's Rule</u>**, determine the limit

$$\lim_{x \to 0} \frac{2\sin(x^2)}{3 - 3\cos(2x)}.$$

(12 marks)

(b) Determine the Taylor expansion of $\sqrt{1 + x^3}$ about $a = 0$, up to and including the term in x^3. Use this Taylor expansion to estimate the value of $\sqrt{1.001}$, giving your answer to 4 decimal places.

(8 marks)

Question 5

Let f be the function defined by

$$f(x) = \frac{3 + x^2}{-x^2 - 2x + 3}.$$

(a) Apply the first derivative test to compute the interval on which f is increasing.

(6 marks)

(b) Determine the horizontal and vertical asymptotes of f if they exist. Justify your answers.

(7 marks)

(c) Determine the critical point(s) of f. Apply relevant derivative tests and discuss the nature of these critical point(s). Show your steps clearly.

(7 marks)

----- END OF PAPER -----